# Лабораторная работа №6

Установка и настройка системы управления базами данных MariaDB

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***дисциплина:*** **Администрирование сетевых подсистем (Lab 6)**

# Цель работы

Целью данной работы является приобретение практических навыков по установке и конфигурированию системы управления базами данных на примере программного обеспечения MariaDB.

```
 mariadb-server              x86_64    3:10.5.29-3.el9_7      appstream    9.7 M
Upgrading:
 selinux-policy              noarch    38.1.65-1.el9_7.1      baseos        37 k
 selinux-policy-targeted     noarch    38.1.65-1.el9_7.1      baseos       6.5 M
Installing dependencies:
 mariadb-common              x86_64    3:10.5.29-3.el9_7      appstream     27 k
 mariadb-errmsg              x86_64    3:10.5.29-3.el9_7      appstream    210 k
 mysql-selinux               noarch    1.0.14-1.el9_6         appstream     36 k
 perl-DBD-MariaDB            x86_64    1.21-17.el9            appstream    149 k
 perl-Sys-Hostname           x86_64    1.23-481.1.el9_6       appstream     15 k
Installing weak dependencies:
 mariadb-backup              x86_64    3:10.5.29-3.el9_7      appstream    6.5 M
 mariadb-gssapi-server       x86_64    3:10.5.29-3.el9_7      appstream     14 k
 mariadb-server-utils        x86_64    3:10.5.29-3.el9_7      appstream    210 k

Transaction Summary
================================================================================
Install  10 Packages
Upgrade   2 Packages

Total download size: 25 M
Downloading Packages:
(1/12): mariadb-common-10.5.29-3.el9_7.x86_64.r  87 kB/s |  27 kB     00:00
(2/12): mariadb-errmsg-10.5.29-3.el9_7.x86_64.r  56 kB/s | 210 kB     00:03
```

Open

my.cnf
/etc

Save

```
#
# This group is read both both by the client and the server
# use it for options that affect everything
#
[client-server]

#
# include all files from the config directory
#
# !includedir /etc/my.cnf.d
```

```
** (gedit:18155): WARNING **: 08:48:46.983: Set document metadata failed: Set
::gedit-position not supported

(gedit:18155): dconf-WARNING **: 08:48:47.002: failed to commit changes to dc
child process “dbus-launch” (No such file or directory)
[root@server etc]# systemctl start mariadb
[root@server etc]# systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/m
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service
m/mariadb.service.
[root@server etc]# ss -tulpen | grep mysql
[root@server etc]# netstat -turpn | prep-w 3306
bash: prep-w: command not found...
[root@server etc]# netstat -turpn | grep 3306
[root@server etc]# netstat -tulpen | grep 3306
tcp6       0      0 :::3306                 :::*                    LISTEN
  18288/mariadbd
[root@server etc]#
```

```
NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] Y
Enabled successfully!
Reloading privilege tables..
 ... Success!


You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] Y
New password:
Re-enter new password:
Sorry, you can't use an empty password here.

New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
 ... Success!


By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n]
```

Examine

Examine

```
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> \h

General information about MariaDB can be found at
http://mariadb.org

List of all client commands:
Note that all text commands must be first on line and end with ';'
?         (\?) Synonym for `help'.
charset   (\C) Switch to another charset. Might be needed for processing binlog with multi-byte char
sets.
clear     (\c) Clear the current input statement.
connect   (\r) Reconnect to the server. Optional arguments are db and host.
delimiter (\d) Set statement delimiter.
edit      (\e) Edit command with $EDITOR.
ego       (\G) Send command to MariaDB server, display result vertically.
exit      (\q) Exit mysql. Same as quit.
go        (\g) Send command to MariaDB server.
help      (\h) Display this help.
nopager   (\n) Disable pager, print to stdout.
notee     (\t) Don't write into outfile.
nowarning (\w) Don't show warnings after every statement.
pager     (\P) Set PAGER [to_pager]. Print the query results via PAGER.
print     (\p) Print current command.
prompt    (\R) Change your mysql prompt.
quit      (\q) Quit mysql.
rehash    (\#) Rebuild completion hash.
sandbox   (\-) Disallow commands that access the file system (except \P without an argument and \e).
source    (\.) Execute an SQL script file. Takes a file name as an argument.
status    (\s) Get status information from the server.
system    (\!) Execute a system shell command.
tee       (\T) Set outfile [to_outfile]. Append everything into given outfile.
use       (\u) Use another database. Takes database name as argument.
warnings  (\W) Show warnings after every statement.

For server side help, type 'help contents'

MariaDB [(none)]>
```

```
(gedit:18481): dconf-WARNING **: 09:10:12.511: failed to commit changes to dconf: Failed to execute
child process “dbus-launch” (No such file or directory)
[root@server my.cnf.d]# systemctl restart mariadb
[root@server my.cnf.d]# gedit utf8.cnf

(gedit:18617): dconf-WARNING **: 09:10:55.372: failed to commit changes to dconf: Failed to execute
child process “dbus-launch” (No such file or directory)

(gedit:18617): dconf-WARNING **: 09:10:55.376: failed to commit changes to dconf: Failed to execute
child process “dbus-launch” (No such file or directory)

(gedit:18617): dconf-WARNING **: 09:10:55.578: failed to commit changes to dconf: Failed to execute
child process “dbus-launch” (No such file or directory)

(gedit:18617): dconf-WARNING **: 09:10:55.579: failed to commit changes to dconf: Failed to execute
child process “dbus-launch” (No such file or directory)

(gedit:18617): dconf-WARNING **: 09:10:55.580: failed to commit changes to dconf: Failed to execute
child process “dbus-launch” (No such file or directory)

** (gedit:18617): WARNING **: 09:11:08.717: Set document metadata failed: Setting attribute metadata
::gedit-position not supported

(gedit:18617): dconf-WARNING **: 09:11:08.766: failed to commit changes to dconf: Failed to execute
child process “dbus-launch” (No such file or directory)
[root@server my.cnf.d]# systemctl restart mariadb
[root@server my.cnf.d]# mysql -u root -p
Enter password:
```

root@server:/etc/my.cnf.d

```
SSL:                    Not in use
Current pager:          stdout
Using outfile:          ''
Using delimiter:        ;
Server:                 MariaDB
Server version:         10.5.29-MariaDB MariaDB Server
Protocol version:       10
Connection:             Localhost via UNIX socket
Server characterset:    latin1
Db     characterset:    latin1
Client characterset:    utf8
Conn.  characterset:    utf8
UNIX socket:            /var/lib/mysql/mysql.sock
Uptime:                 12 min 24 sec

Threads: 1  Questions: 4  Slow queries: 0  Opens: 17  Open tables: 10  Queries per second avg: 0.005
--------------

MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> USE addressbook;
Database changed
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.003 sec)

MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.097 sec)

MariaDB [addressbook]> NSERT INTO city(name,city) VALUES ('Иванов','Москва');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near 'NSERT INTO city(name,city) VALUES ('Иванов'
,'Москва')' at line 1
MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов','Москва');
Query OK, 1 row affected (0.026 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Петров','Сочи');
Query OK, 1 row affected (0.014 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Сидоров','Дубна');
Query OK, 1 row affected (0.013 sec)

MariaDB [addressbook]>
```

```
MariaDB [addressbook]> CREATE USER user@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.012 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO frank@'%';
ERROR 1133 (28000): Can't find any matching row in the user table
MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user'%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near ''%'' at line 1
MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user"%";
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near '"%"' at line 1
MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user"%";
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near '"%"' at line 1
MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> DESCRIBE city;
+-------+-------------+------+-----+---------+-------+
| Field | Type        | Null | Key | Default | Extra |
+-------+-------------+------+-----+---------+-------+
| name  | varchar(40) | YES  |     | NULL    |       |
| city  | varchar(40) | YES  |     | NULL    |       |
+-------+-------------+------+-----+---------+-------+
2 rows in set (0.002 sec)

MariaDB [addressbook]> quit
Bye
[root@server my.cnf.d]# mysqlshow -u root -p
Enter password:
+--------------------+
|     Databases      |
+--------------------+
| addressbook        |
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
[root@server my.cnf.d]#
```

```
MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user'%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near ''%'' at line 1
MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user"%";
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near '"%"' at line 1
MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE ON addressbook.* TO user"%";
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your
MariaDB server version for the right syntax to use near '"%"' at line 1
MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> DESCRIBE city;
+-------+-------------+------+-----+---------+-------+
| Field | Type        | Null | Key | Default | Extra |
+-------+-------------+------+-----+---------+-------+
| name  | varchar(40) | YES  |     | NULL    |       |
| city  | varchar(40) | YES  |     | NULL    |       |
+-------+-------------+------+-----+---------+-------+
2 rows in set (0.002 sec)

MariaDB [addressbook]> quit
Bye
[root@server my.cnf.d]# mysqlshow -u root -p
Enter password:
+--------------------+
|     Databases      |
+--------------------+
| addressbook        |
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
[root@server my.cnf.d]# mysqlshow -u root -p addressbook
Enter password:
Database: addressbook
+--------+
| Tables |
+--------+
| city   |
+--------+
[root@server my.cnf.d]# mysqlshow -u user -p addressbook
```

# *Вывод*

В ходе выполнения лабораторной работы были приобретены практические навыки по установке и конфигурированию системы управления базами данных на примере программного обеспечения MariaDB.

*Спасибо за внимание!*